

Lesson 3

Business Process Design Issues

Lesson objectives

- Describe basic business process concepts
- Monitor and manage business processes
- Set performance-related options on the check-in screen
- Manually run a business process and track results
- List guidelines and best practices for business process design and efficient processing
- Describe the thread monitor function
- Explain the guidelines for large file handling

Business Process Development and Maintenance

Developing and maintaining business process models involve the following tasks:

Analysis and Design

- Planning the most efficient processes to achieve the business objectives.
- Determining the Sterling B2B Integrator components needed to complete the tasks in the processes.

Coding and Testing

- Configuring business process model components
- Creating business process models
- Testing process models
- Implementing the business process models in Sterling B2B Integrator

Maintenance

- Modifying business process models as changing needs require them to grow, shrink, change, merge, and split
- Keeping records of the process models you create, use, and modify, for reference

Business Process Management Tasks

The business process management tasks are:

Create a business process

Validate a business process

Check In a business process

Start and stop a business process

Archive a business process

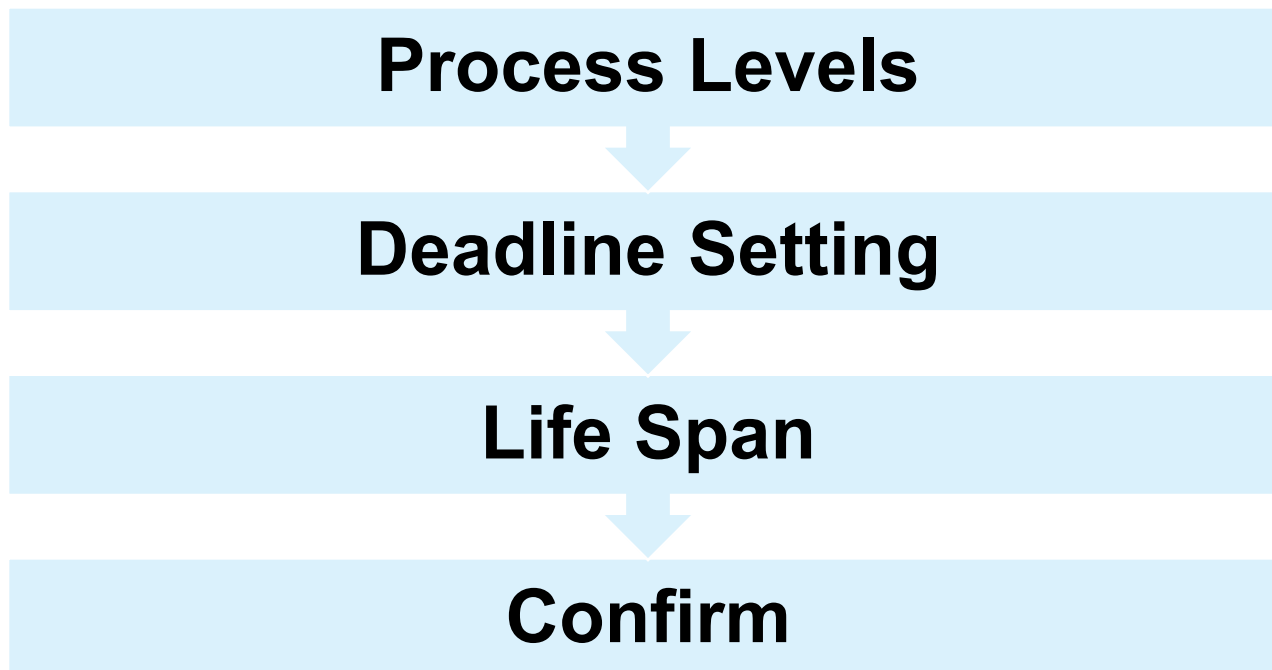
Restore a business process

Delete a business process

Monitor a business process

Checking in a Business Process

While checking in a business process, the following sections on the Editor page pertain to system performance:



Best Practices

The following best practices will help to ensure optimum performance of the business processes:

- Name versions meaningfully
- Retain service names in Graphical Process Models
- Reusability
- Use predefined business processes
- Keep subprocesses manageable
- Invoke subprocesses asynchronously when appropriate
- Make modifications incrementally

Best Practices

(contd..)

- Use appropriate components
- Deprecation tool
- Run batch transactions where possible
- Test for execution speed and modify business process models as needed
- Limit steps in a business process model
- Use schedules to manage workload
- Use Wait and Sleep services appropriately

Best Practices

(contd..)

- Limit calls to the file system
- Use BPML All activity sparingly
- Manage process data
- Only pass child processes what they need
- Obtain process information in OnFault layer
- Ensure that relevant onFaults are present
- Develop service as last resort

Use Specified Path in XPath Searches

- When making an XPath reference to data, using a specified path is more efficient than a "/" search
- Using a "/" search finds every possible node that matches the selection criteria, regardless of how deeply nested it is.
- The XPath expression language is fairly powerful. Unfortunately, XPath evaluation is expensive – particularly when dealing with complex predicate expressions.



Write XPath statements using absolute paths. For example, write `PurchaseOrder/text` instead of `/ProcessData/PurchaseOrder/text()`.

The Invoke Business Process Service

The Invoke Business Process Service is used to start an instance of a subprocess (child) from within the current instance of the business process (parent).

To implement the Invoke Business Process Service, complete the following tasks:

1. Create an Invoke Business Process Service configuration.
2. Create and enable a business process that includes the Invoke Business Process Service.
3. Test the business process and the service.
4. Run the business process.

Thread Monitor

- You can monitor threads related to business processes in Sterling B2B Integrator using the Thread Monitor.
- The Thread Monitor lists all threads currently running in the Activity engine.
- This information may be useful for troubleshooting business processes that appear to have stopped or are hung.

Guidelines: Large File Handling

- To ensure efficient processing of large files, you must configure the business process models to prevent unnecessary overloading of the database.
- Using the For Each Document service along with an Invoke BP service in a loop can be a source of production down issues when handling files of any size, but particularly for large files.



Guidelines: Large File Handling

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- To avoid this problem, structure the business process model as one of the following methods:

Method 1:

In the business process model, include:

- An Assign activity with a message_to_child element that selectively passes the correct document to the next cycle in the loop
- A Release Service placed to clear out unnecessary document copies

Method 2:

In the business process model, include:

- One or more assign elements to message_to_child in the Message To Service of the Invoke Business Process service, to specify the correct document as an output element.
- Need not use the Release service if this configuration is used.



Exercise

1. Persistence level business process
2. Document Storage Settings
3. Monitor invoke mode configuration
4. Using message tags
5. Sleep Service

Completed lesson objectives

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Questions

